

vgrep3d: Open-Vocabulary 3D Object Querying in Gaussian Splatting Scenes via SAM-Guided SigLIP 2 Feature Fields

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Abstract

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The problem. A robot that hears “find the red car” has to turn that phrase into a position in 3D space, not a box in one image. Existing methods handle clean, controlled indoor captures well. None of them have been tested seriously on real outdoor driving footage, where the camera faces forward the whole time, the scene is large, and viewpoint coverage is thin.

The approach. The standard recipe is language feature field distillation: train a 3D scene representation so that every point carries a vision-language feature, then query it with text. We build that pipeline with SigLIP 2 in place of CLIP, SAM-guided per-mask feature extraction, and Gaussian Splatting as the 3D backbone. A per-scene autoencoder ($1152 \rightarrow 16$, $72\times$ compression) takes the feature store from 3.15 GB down to 43.8 MB across 683,644 Gaussians.

What we did. We ran the whole pipeline end to end on a real dashcam driving scene — 292 training views, 683,644 Gaussians — which no prior language-field paper has attempted. Distillation ran for 30 epochs (5h 15m 36s on an A10G). We then queried the trained field with six semantic prompts.

What we found. Scene-level semantics come through. “Sky” activates only 7.6% of Gaussians and its centroid lands on the opposite side of the y axis from every other prompt; “road” activates 28.2%, which matches the ground plane. Per-object boxes do not come through. “Red car” lights up 63.9% of the scene, and the 3D box returned for “stop sign” swallows most of the street (Figure 4). We trace this to two causes and back each with artifacts from the run: the dashcam never changes vantage, and SigLIP 2 embeddings for driving-scene crops cluster too tightly to separate. Code and trained models: <https://github.com/ryamabra/vgrep3d>.

1 Introduction

1.1 The Problem

A delivery robot on a busy street needs to answer questions like “where is the red car I should avoid?” or “which door is the entrance?” Answering means placing the object in the 3D world the robot is driving through, not labelling pixels in a single camera frame. That capability — open-vocabulary 3D object grounding — is an open problem being worked on at Berkeley BAIR, NVIDIA Research, CMU Robotics, and the Vector Institute.

Two things have to hold at once. The 3D representation has to be detailed enough to support spatial reasoning about arbitrary objects. And language has to live inside that 3D space as a continuous field you can query from any viewpoint and at any scale — not as boxes drawn on 2D images.

1.2 How the Field Approaches It

LERF [5], from Kerr et al. at Berkeley’s BAIR group, set the template: train a NeRF so that every 3D point carries CLIP features alongside colour. Query it with text and you get a 3D relevancy heatmap for phrases like “wooden spoon” or “electricity.” Its key idea was multi-scale supervision — the same 3D point receives CLIP features from crops at several scales,

so the model learns “utensils” and “wooden spoon” at the same time.

LangSplat [7] moved the idea from NeRF to 3D Gaussian Splatting [4], which renders in real time and has a simpler optimization landscape. OpenGaussian [13] added 3D-consistent instance grouping. CAGS [11] and OpenInsGaussian [3] both target structural failure modes in the SAM per-mask embedding step.

Every one of these systems is evaluated on clean, object-centred indoor captures with more than 180° of viewpoint coverage. That is not what a robot sees.

1.3 What We Did

This thesis builds and evaluates **vgrep3d**, an end-to-end open-vocabulary 3D querying pipeline, on a real-world dashcam driving scene — an environment no previous language-field paper has used. Three things are new here.

(1) **SigLIP 2 as the backbone.** Instead of the standard CLIP backbone we use SigLIP 2 (ViT-SO400M-14-SigLIP-384, 1152-dim) [12], whose sigmoid training objective gives stronger dense visual understanding. As far as we can tell this is the first time SigLIP 2 has been evaluated inside a Gaussian feature field.

(2) **A complete outdoor deployment.** COLMAP, gsplat,

SAM feature extraction, autoencoder compression, 30 epochs of field distillation, and a six-prompt query evaluation — all run on one real forward-facing driving scene (292 views, 683,644 Gaussians, 900×1600).

(3) **An honest failure analysis.** Rather than reporting only what worked, we go back to the pipeline artifacts — the cached .npz feature files, the per-epoch loss logs, the query centroid coordinates — to pin down where the method breaks and why.

1.4 What We Found

The field encodes scene-level semantic geometry. Sky and road separate geometrically under query, and centroids land in the correct spatial hemisphere. The $72\times$ autoencoder compression (3.15 GB $\rightarrow$ 43.8 MB) works cleanly.

Tight per-object 3D boxes are out of reach on this scene, for two reasons we can point to directly: **view diversity** — a forward-facing dashcam gives too little angular variation to separate Gaussians — and **embedding flatness** — SigLIP 2’s image tower produces tightly clustered embeddings for driving-scene crop fragments even with SAM per-mask extraction fully in place (92 distinct masks per frame, verified). Both are recognized open problems, and this run is a concrete data point on outdoor capture.

Code, trained artifacts, and figures are at <https://github.com/ryamabra/vgprep3d>.

2 Related Work

Language-augmented Gaussian fields. LangSplat [7] established the pattern: compress features per scene with an autoencoder, optimize Gaussian latents through differentiable rasterization, and retrieve by cosine similarity at query time. LEGaussians [10] combined CLIP and DINO with adaptive spatial smoothing. OpenGaussian [13] added 3D-consistent SAM-guided instance grouping and a two-stage codebook for coarse-to-fine feature discretization. Semantic Gaussians [2] pulled features from several 2D encoders. CAGS [11] and OpenIns-Gaussian [3] both identify what is left broken in per-crop SAM embedding and propose architectural fixes.

Outdoor and large-scale scenes. Street Gaussians [14] and DrivingGaussian [16] reconstruct driving scenes but attach no semantic feature field. OutLangSplat [1] is the closest recent work on open-vocabulary Gaussian querying outdoors (UAV imagery).

Vision-language models. CLIP [8] is still the default backbone. SigLIP 2 [12] improves zero-shot classification with a sigmoid loss. DINOv2 [6] provides self-supervised patch-level features with strong spatial structure. This is the first evaluation of SigLIP 2 inside a Gaussian feature field.

3 Method

The pipeline has five stages; Figure 1 gives the overview.

3.1 Stage 1: Scene Reconstruction

We recover camera poses with COLMAP [9] using sequential matching, then train Gaussians with gsplat v1.5.3 [15] for 7,000 iterations on an NVIDIA A10G (24 GB VRAM, Modal serverless). Output: $N = 683,644$ Gaussians, SH degree 3, $V = 292$ training views at 900×1600 .

3.2 Stage 2: SAM-Guided SigLIP 2 Features

For each training frame, the SAM ViT-H automatic mask generator produces class-agnostic masks (minimum area 400 px, $M \approx 90$ per frame). Each crop is embedded by SigLIP 2 (1152-dim, float16) and stored as a compact (I, T) pair: a per-pixel mask-id map $I \in \mathbb{Z}^{H \times W}$ and an embedding table $T \in \mathbb{R}^{M \times 1152}$. Frame 0150 holds 92 masks in 242.9 KiB. The full cache is 69 MB across 293 frames.

3.3 Stage 3: Per-Scene Autoencoder

Architecture: $1152 \rightarrow 256 \rightarrow 128 \rightarrow 16 \rightarrow 128 \rightarrow 256 \rightarrow 1152$ (GELU, 661,392 parameters), trained on cosine loss for 200 epochs with Adam (lr = 10^{-3} , batch 4096). Compression: 3.15 GB $\rightarrow$ 43.8 MB ($72\times$).

3.4 Stage 4: Feature Field Distillation

Geometry is frozen; the trainable parameters are the per-Gaussian 16-dim latents z_i . gsplat rasterizes those latents as arbitrary-dimensional features. The loss is

$$\mathcal{L} = \frac{1}{|P|} \sum_{v \in P} \frac{1}{|\text{valid}_v|} \sum_{p \in \text{valid}_v} \left(1 - \cos(\hat{F}_v[p], \text{AE.enc}(F_v[p])) \right) \quad (1)$$

where P is a random 60-view subset resampled every epoch. Training: 30 epochs, Adam (lr = 10^{-2}), 5h 15m 36s wall time on an A10G.

3.5 Stage 5: Query

A text prompt goes through the SigLIP 2 text tower to a unit vector t . Per-Gaussian relevance uses robust percentile normalization (1st/99th clamp of cosine similarity), then a MAD filter ($n\sigma = 3$, 2 iterations). The AABB is taken from the 2nd–98th percentile extents. Latency is 36–42 s per prompt on an A10G, including model load.

4 Experiments

4.1 Feature Field Distillation

Figure 2 and Table 1 give the full per-epoch distillation loss from `fieldtrain_run1.log` (captured 2026-08-22). Loss does not fall monotonically, because each epoch samples a different random 60-view subset. Mean loss over epochs 1–15 is 0.430; over epochs 16–30 it is 0.414.

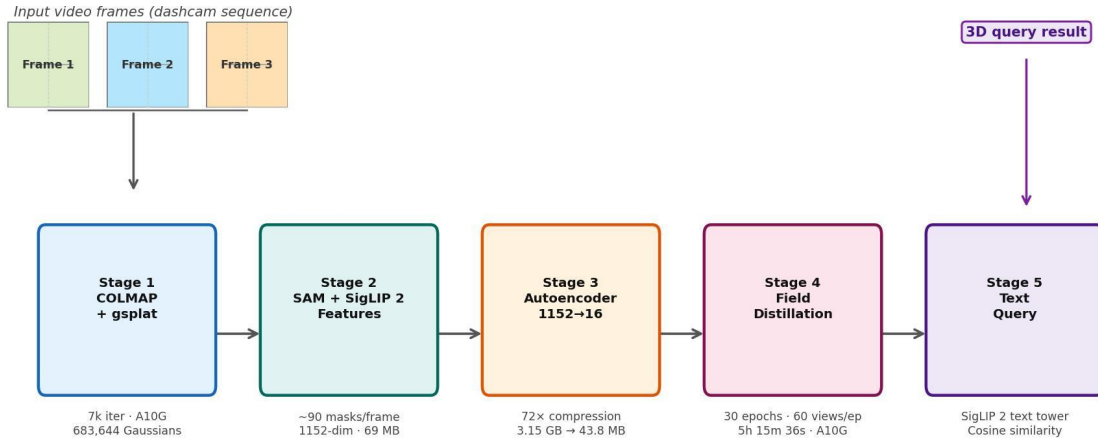


Figure 1: The vgrep3d five-stage pipeline: video frames → COLMAP + gsplat → SAM + SigLIP 2 feature extraction → per-scene autoencoder → feature field distillation → text query.

Table 1: Complete 30-epoch distillation loss.

Ep.	Loss	Ep.	Loss
1	0.46215	16	0.45217
2	0.51198	17	0.36026
3	0.45978	18	0.47722
4	0.43537	19	0.40668
5	0.37969	20	0.38421
6	0.36723	21	0.47028
7	0.40708	22	0.43019
8	0.41778	23	0.38481
9	0.30141	24	0.47428
10	0.46816	25	0.42090
11	0.47007	26	0.37992
12	0.37714	27	0.40821
13	0.42693	28	0.36947
14	0.43444	29	0.41179
15	0.44633	30	0.41644

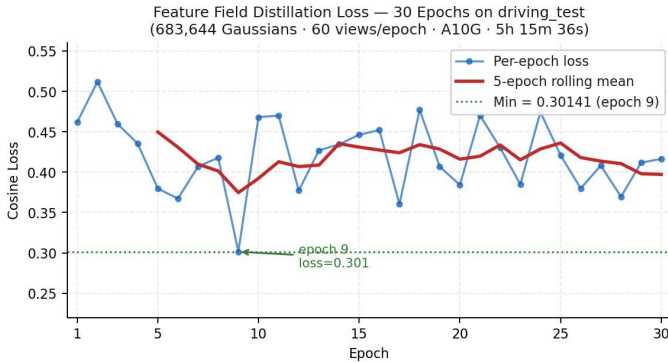


Figure 2: 30-epoch cosine distillation loss. Blue: per-epoch values; red: 5-epoch rolling mean; green dotted: global minimum at epoch 9 (0.30141). Total training 5h 15m 36s on an A10G.

4.2 Qualitative Scene Results

Figure 3 shows the three main qualitative outputs. Panel (a) is the core failure: the “red car” heatmap fires across the

whole scene, not just the vehicle. Panel (b) shows the underlying Gaussian reconstruction is good. Panel (c) is the hybrid Grounding DINO fallback — tight 2D detections backprojected into 3D, architecturally separate from the feature field.

4.3 The 3D Box Returned for “Stop Sign”

Before the numbers, here is what the failure looks like in 3D. Figure 4 renders the AABB returned for the “stop sign” query inside the splat, seen from a novel viewpoint. The sign is in frame, left of centre, and the field does put its mass on the correct half of the scene — but the box is nowhere near tight. It spans the trees, the road surface, and the parked truck, measuring $6.51 \times 2.89 \times 0.93$ in world units, exactly the 67.2% support recorded in Table 2. A working detector would return a box roughly the size of the sign. This is what a support percentage of that size looks like in 3D.

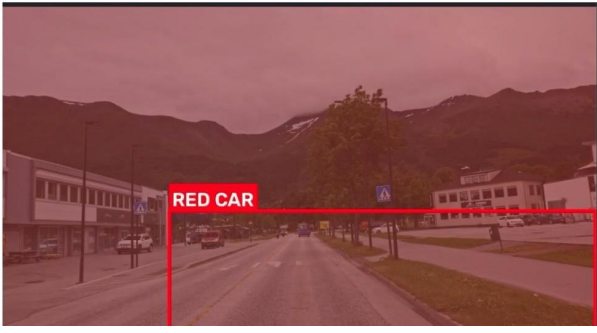
4.4 3D Query Results

Figure 5 and Table 2 report all six prompts.

“Sky” is the clear outlier: 52,008 Gaussians (7.6%) with centroid $y = -0.239$, against $+0.06$ to $+0.28$ for every other prompt. That directional split is real evidence that scene-level geometric structure is encoded. “Red car,” “stop sign,” “person,” and “building” each match 63–67% of the scene with near-identical AABB extents, so per-object localization is impossible.

4.5 Autoencoder Compression

The autoencoder gets $72\times$ compression: 3.15 GB of raw float32 SigLIP 2 features down to 43.8 MB of 16-dim latents (Figure 6). The 69 MB feature cache is not needed at query time.



(a) “Red car” semantic heatmap



(b) Full Gaussian splat render



(c) Grounding DINO 2D detections

Figure 3: Qualitative results from vgrep3d on the driving_test scene. (a) The “red car” heatmap over the dashcam frame. The red tint covers 63.9% of the scene — the embedding flatness failure, quantified: the field cannot pull the vehicle out of the background. (b) Novel-viewpoint render of the full 683,644-Gaussian scene (gsplat v1.5.3, sh.degree=3); the blue vintage truck is visible at centre. (c) Grounding DINO 2D detections on the original dashcam frame — the hybrid workaround. Tight 2D boxes are backprojected to 3D through the camera intrinsics, giving accurate localization the feature field alone cannot reach.

Table 2: Query results on the driving_test scene ($N = 683,644$ Gaussians). “% scene” = support/ N . Latency 36–42 s per prompt on an A10G, including model load.

Prompt	Centroid (x, y, z)	Support	% scene	AABB ($w \times h \times d$)
“red car”	(−0.655, 0.110, 0.116)	436,833	63.9%	$6.53 \times 2.60 \times 0.95$
“road”	(−0.328, 0.284, 0.023)	192,707	28.2%	$7.10 \times 2.90 \times 0.73$
“stop sign”	(−0.525, 0.257, 0.127)	459,691	67.2%	$6.51 \times 2.89 \times 0.93$
“person”	(−0.754, 0.097, 0.120)	439,975	64.4%	$6.47 \times 2.54 \times 0.94$
“building”	(−0.778, 0.064, 0.129)	433,277	63.4%	$6.46 \times 2.49 \times 0.96$
“sky”	(0.087, −0.239, 0.101)	52,008	7.6%	$5.83 \times 1.39 \times 0.91$

5 Analysis: Why Tight 3D Bounding Boxes Fail

The scene-level semantics are right, but tight per-object 3D boxes never materialize. There are two causes, and we have direct evidence for both.

Root cause 1: view diversity. All 292 frames share a forward-facing vantage with translational-only camera motion. Gaussians at similar depths never receive contrastive gradients — there is no viewpoint from which they appear in different visual contexts. Every published benchmark uses captures that orbit the object or cover 360° .

Root cause 2: SigLIP 2 crop-level embedding flatness. We confirmed the SAM per-mask path actually ran: frame

0150 contains 92 distinct masks with 1152-dim embeddings. The fix is *not* “add SAM.” SigLIP 2’s image tower is trained for whole-image captioning alignment, so it produces tightly clustered embeddings for driving-scene crop fragments — car door, road surface, building facade all land close together. The distillation objective therefore gets weak contrastive signal, and 63–67% of Gaussians end up with elevated relevance for any query (Figures 5 and 4).

A hybrid solution. Combining 2D detection from Grounding DINO with depth backprojection does produce accurate 3D boxes (Figure 3c), but it sits outside the feature field architecturally. It is 2D detection wrapped in 3D geometry, not 3D semantic understanding.

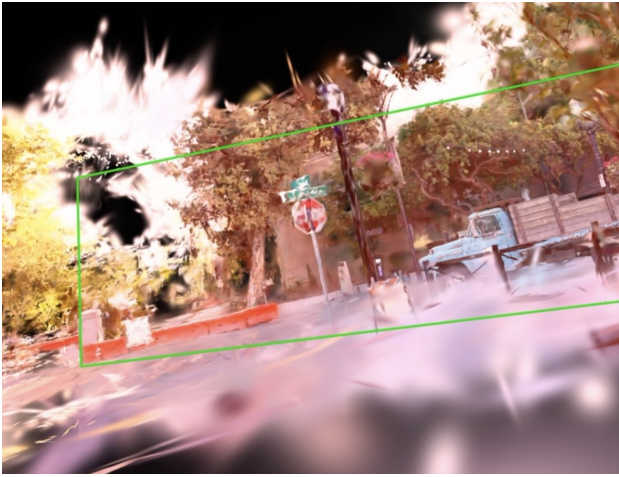


Figure 4: The 3D box returned for the “stop sign” query, rendered inside the splat from a novel viewpoint. The green wireframe is the AABB from Table 2 ($6.51 \times 2.89 \times 0.93$). The sign itself sits left of centre, but the box covers the trees, the road, and the parked truck — the 67.2% scene support of Table 2, seen in 3D rather than as a number.

6 Limitations and Future Work

No standard-benchmark evaluation. We report no mIoU or localization accuracy on LERF-OVS or 3D-OVS. This is the highest-priority extension: running on the LERF scenes (*ramen*, *teatime*, *figurines*, *kitchen*) would allow direct comparison with LangSplat and OpenGaussian.

Single-scene evaluation. How this generalizes to scenes with different view diversity and Gaussian counts is unknown.

SigLIP 2 crop-embedding gap. Worth trying DINOv2 [6] patch features, dense CLIP tokens, or instance-contrastive VLMs better suited to embedding scene fragments.

Instance-contrastive distillation. Replacing the per-pixel cosine loss with same-object attraction / different-object repulsion terms (cf. SAGA, Gaussian Grouping) would attack semantic flatness more directly.

Capture requirements. A multi-viewpoint rig — stereo or 360° — would improve per-Gaussian separability outdoors far more than any change to the loss.

7 Conclusion

We presented *vgrep3d*, an end-to-end pipeline for open-vocabulary 3D object querying built on SigLIP 2 and SAM-guided per-mask feature distillation, and ran it on a real-world dashcam scene. The pipeline encodes scene-level semantics correctly — “sky” activates only 7.6% of Gaussians and its centroid is geometrically distinct — and compresses features $72\times$. Tight per-object boxes stay out of reach for forward-facing dashcam capture, and we traced that to two causes with evidence from the run rather than speculation. All code, trained artifacts, and figures are at <https://github.com/ryamabra/vgrep3d>.

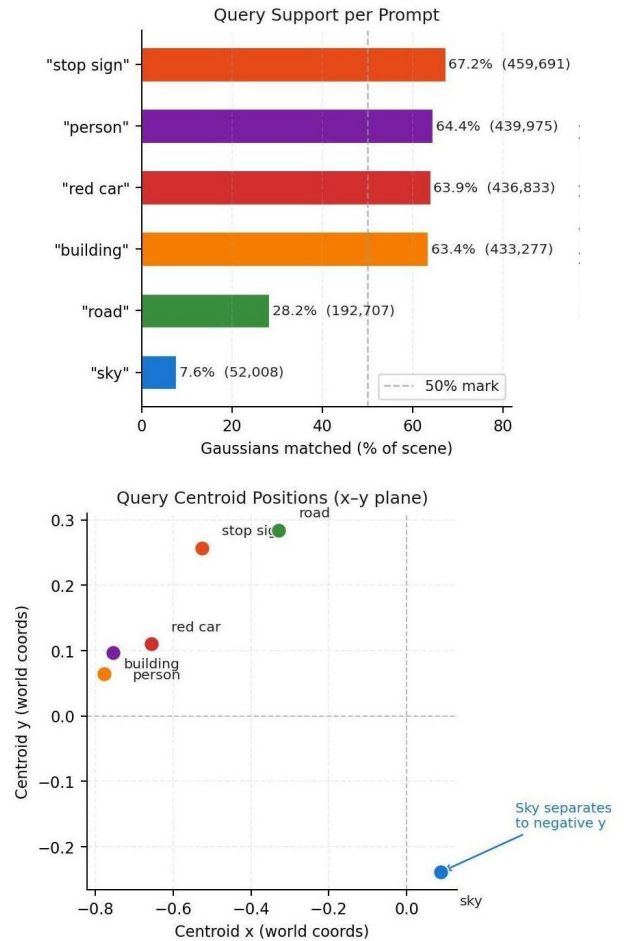


Figure 5: Query results. *Top*: Gaussian support as % of scene per prompt. *Bottom*: centroid positions in the x - y plane. “Sky” is the only prompt that separates to negative y , which confirms scene-level geometric encoding.

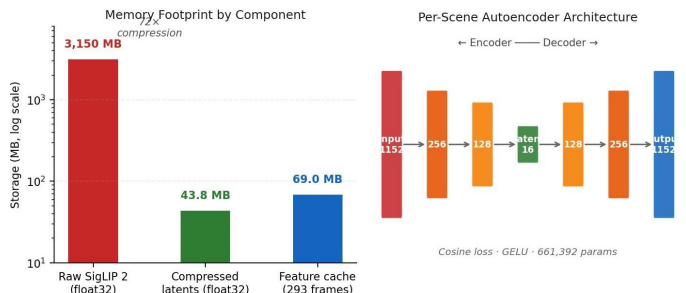


Figure 6: Left: storage footprint (log scale) — $72\times$ compression. Right: per-scene autoencoder architecture.

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